

P2-S2-2016

1. Resuelva el siguiente sistema de ecuaciones.

$$\begin{cases} 2x + 7y - 3z - 2w = 5 \\ x + 3y - z - w = 2 \\ x + 2y - w = 1. \end{cases}$$

$$\left(\begin{array}{cccc|c} 2 & 7 & -3 & -2 & 5 \\ 1 & 3 & -1 & -1 & 2 \\ 1 & 2 & 0 & -1 & 1 \end{array} \right)$$

$$\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

Pivotes

$$\left(\begin{array}{cccc|c} 2 & 7 & -3 & -2 & 5 \\ 1 & 3 & -1 & -1 & 2 \\ 1 & 2 & 0 & -1 & 1 \end{array} \right)$$

$$F2 \leftrightarrow F3 \left(\begin{array}{cccc|c} 2 & 7 & -3 & -2 & 5 \\ 1 & 2 & 0 & -1 & 1 \\ 1 & 3 & -1 & -1 & 2 \end{array} \right)$$

$$\frac{1}{2} \cdot F1 \left(\begin{array}{cccc|c} 1 & \frac{7}{2} & -\frac{3}{2} & -1 & \frac{5}{2} \\ 1 & 2 & 0 & -1 & 1 \\ 1 & 3 & -1 & -1 & 2 \end{array} \right)$$

$$-F1 + F2 \left(\begin{array}{cccc|c} 1 & \frac{7}{2} & -\frac{3}{2} & -1 & \frac{5}{2} \\ 0 & -\frac{3}{2} & \frac{3}{2} & 0 & -\frac{3}{2} \\ 1 & 3 & -1 & -1 & 2 \end{array} \right)$$

$$-F_1 + \widetilde{F}_3 \quad \left(\begin{array}{cccc|c} 1 & \frac{7}{2} & \frac{-3}{2} & -1 & \frac{5}{2} \\ 0 & \frac{-3}{2} & \frac{3}{2} & 0 & \frac{-3}{2} \\ 0 & \frac{2}{3} & \frac{7}{2} & 0 & \frac{-1}{2} \end{array} \right)$$

$$\frac{-2}{3} \cdot \widetilde{F}_2 \quad \left(\begin{array}{cccc|c} 1 & \frac{7}{2} & \frac{-3}{2} & -1 & \frac{5}{2} \\ 0 & 1 & -1 & 0 & 1 \\ 0 & \frac{2}{3} & \frac{7}{2} & 0 & \frac{-1}{2} \end{array} \right)$$

$$-\frac{7}{2} \cdot F_2 + \widetilde{F}_1 \quad \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & -1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & \frac{2}{3} & \frac{7}{2} & 0 & \frac{-1}{2} \end{array} \right)$$

$$-\frac{2}{3} \cdot F_2 + \widetilde{F}_3 \quad \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & -1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & \frac{7}{6} & 0 & \frac{-7}{6} \end{array} \right)$$

$$\frac{6}{7} \cdot \widetilde{F}_3 \quad \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & -1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \end{array} \right)$$

$$F_3 + \widetilde{F}_1 \quad \left(\begin{array}{cccc|c} 1 & 0 & 2 & -1 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{array} \right)$$

$$-2 \cdot F_3 + \widetilde{F}_1 \quad \begin{array}{c} x \quad y \quad z \quad w \\ \left(\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{array} \right) \end{array}$$

$$x - w = 1 \rightarrow x = 1 + w$$

$$y = 0$$

$$z = -1$$

$$x - w = 1 \rightarrow x = 1 + w$$

$$y = 0$$

$$z = -1$$

$$S = \{(1+w, 0, -1, w)\}$$

5. Considere las matrices A y X de dimensiones 3×3 y la matriz B de dimensión 2×3 , para las cuales está definida la igualdad

$$XA + B^t B = 2X.$$

a) Utilice álgebra matricial para despejar X de la ecuación anterior. (3 puntos)

$$XA + B^t B = 2X$$

$$B^t B = 2X - XA$$

$$A = AI$$

$$B^t B = 2XI - XA$$

$$\begin{aligned} &2X - AX \\ &(2I - A)X \end{aligned}$$

$$B^t B = X(2I - A)$$

$$X = (B^t B) \cdot (2I - A)^{-1}$$

b) Determine X de forma explícita, si $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$ y $B = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}$. $X = (B^t B) \cdot (2I - A)^{-1}$
(4 puntos)

$$B = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}, \quad B^t = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{pmatrix}$$

$$B^t B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}$$

$$B^T \cdot B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1+1 & 0-2 & 1+2 \\ 0-2 & 0+2 & 1-4 \\ 1+2 & 0-4 & 1+4 \end{pmatrix}$$

$$B^T \cdot B = \begin{pmatrix} 2 & -2 & 3 \\ -2 & 2 & -3 \\ 3 & -4 & 5 \end{pmatrix}$$

$$2I - A, \quad A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2 \cdot I = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$2I - A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$$

$$2I - A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 4 & 3 \\ 1 & -1 & 0 \end{pmatrix}$$

$$2I - A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 4 & 3 \\ 1 & -1 & 0 \end{pmatrix}$$

$$(2I - A)^{-1} = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 4 & 3 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$F_2 \leftrightarrow \widetilde{F_3} \quad \left(\begin{array}{ccc|ccc} \boxed{1} & 0 & 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-1F_1 + \widetilde{F_2} \quad \left(\begin{array}{ccc|ccc} \boxed{1} & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-1 \cdot \widetilde{F_2} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & \boxed{1} & 1 & 1 & 0 & -1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-4F_2 + \widetilde{F_3} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & \boxed{1} & 1 & 1 & 0 & -1 \\ 0 & 0 & -1 & -4 & 1 & 4 \end{array} \right)$$

$$-1 \cdot \widetilde{F_3} \quad \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 0 & \boxed{1} & 4 & -1 & -4 \end{array} \right)$$

$$\begin{array}{l} -F_3 + \widetilde{F_1} \\ -F_3 + \widetilde{F_2} \end{array} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 4 \\ 0 & 1 & 0 & -3 & 1 & 3 \\ 0 & 0 & 1 & 4 & -1 & -4 \end{array} \right)$$

$$(B^T, B), (2I - A)^{-1}$$

$$\begin{pmatrix} 2 & -2 & 3 \\ -2 & 4 & -3 \\ 3 & -4 & 5 \end{pmatrix} \times \begin{pmatrix} -3 & 1 & 4 \\ -3 & 1 & 3 \\ 4 & -1 & -4 \end{pmatrix}$$

$$\begin{pmatrix} -6 + 6 + 12 & 2 - 2 - 3 & 8 - 6 - 12 \\ 6 - 12 - 12 & -2 + 4 + 3 & -8 + 12 + 12 \\ -9 + 12 + 20 & 3 - 4 - 5 & 12 - 12 - 20 \end{pmatrix}$$

$$\begin{pmatrix} 12 & 3 & -10 \\ -18 & 5 & 16 \\ 23 & -6 & -20 \end{pmatrix}$$

6. Determine el número complejo z que satisface simultáneamente las condiciones siguientes: (4 puntos)

$$\begin{cases} |z-1|=2 \\ \text{Arg}(z-2-i) = -\frac{\pi}{2} \end{cases} \rightarrow \pm \frac{\pi}{2} \begin{matrix} -\frac{\pi}{2} & a=0 & b<0 \\ \frac{\pi}{2} & a=0 & b>0 \end{matrix}$$

$$z = a + bi$$

$$\begin{aligned} a + bi - 2 - i \\ (a-2) + (b-1)i \quad a=0 \quad b<0 \\ a \quad b \end{aligned}$$

$$\begin{aligned} a-2=0 \quad b-1<0 \\ a=2 \quad b<1 \end{aligned}$$

$$|z-1|=2$$

$$|a+bi-1|=2$$

$$|(a-1)+(b)i|=2$$

$$(a-1)^2 + b^2 = 2^2 \quad a=2$$

$$(2-1)^2 + b^2 = 4$$

$$1 + b^2 = 4$$

$$b^2 = 4 - 1$$

$$b^2 = 3$$

$$b = \pm \sqrt{3}$$

$$b = -\sqrt{3}$$

$$a=2 \quad b=-\sqrt{3}$$

$$\boxed{R/a - \sqrt{3}i}$$

→ Rectangular

8. Expresa $(-1 - i)^i$ en la forma $a + bi$, con a y b números reales.

$$z = (-1 - i)^i$$

$$\ln(z) = \ln(-1 - i)^i$$

$$= i \ln(-1 - i)$$

$$i \ln(\sqrt{2} \cdot e^{-\frac{3\pi}{4}i})$$

$$i \ln(\sqrt{2}) + i \ln(e^{-\frac{3\pi}{4}i})$$

$$\ln(\sqrt{2})i + \frac{-3\pi}{4} i^2$$

$$\ln(\sqrt{2})i + \frac{3\pi}{4}$$

$$e^{\ln(\sqrt{2})i} + e^{\frac{3\pi}{4}}$$

$$(\cos(\ln(\sqrt{2})) + i \sin(\ln(\sqrt{2}))) + e^{\frac{3\pi}{4}}$$

$$\{ \cos(\ln(\sqrt{2})) + \sin(\ln(\sqrt{2}))i \} \cdot e^{\frac{3\pi}{4}}$$

$$\cos(\ln(\sqrt{2})) \cdot e^{\frac{3\pi}{4}} + \sin(\ln(\sqrt{2})) \cdot e^{\frac{3\pi}{4}} i$$

$$\boxed{9.92 + 3.58i}$$

$$\ln(x)^n = n \ln(x)$$

$$a = -1 \quad b = -1 \quad i\theta$$

$$-1 - i \rightarrow re$$

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{-1}{-1}\right) - \pi$$

$$= \frac{-3\pi}{4}$$

$$\sqrt{2} \cdot e^{-\frac{3\pi}{4}i}$$

$$re^{i\theta} = r(\cos(\theta) + i \sin(\theta))$$

P2-S2-2018

#2 Si se tiene que $w = \frac{2-5ai}{1+2i}$, determine todos los valores para el número real a , de tal forma que $\text{Im}(w) \neq 0$. (4 puntos)

$$\frac{2-5ai}{1+2i} \cdot \frac{1-2i}{1-2i}$$

$$\frac{(2-5ai)(1-2i)}{1^2 - (2i)^2}$$

$$\begin{aligned} 1 \cdot 1 &= 1 \\ + 4 &= 5 \\ 1+4 &= 5 \end{aligned}$$

$$\frac{2-7i-5ai+10ai^2}{5}$$

$$2-7i-5ai-10a$$

$$\frac{(2-10a) + (-7-5a)i}{5}$$

$$\frac{2-10a}{5} + \frac{-7-5a}{5}i$$

$$\text{Im}(z) \neq 0$$

$$\frac{-7-5a}{5} \neq 0$$

$$-7-5a \neq 0$$

$$-5a \neq 7$$

$$\boxed{a \neq \frac{-7}{5}}$$